1 Unsupervised learning and clustering

Supervised vs. unsupervised learning

- Supervised: we observe predictors and a response, $\{(\mathbf{x}^{(i)}, y^{(i)})\}_{i=1}^{n}$, and learn f such that $y \approx f(\mathbf{x})$
- Unsupervised: we observe only $\mathbf{x}^{(i)} \in \mathbb{R}^p$ for $i = 1, \dots, n$, collected in the $n \times p$ data matrix $\mathbf{X}$
- There is no response y, so there is no "correct answer" to compare against, no test error, and no cross-validation of prediction accuracy
- The goal is to discover structure: subgroups of observations, relationships among variables, low-dimensional summaries

Clustering: the goal

- Partition the n observations into K groups $C_1, \dots, C_K$ so that observations within a group are similar and observations in different groups are different
- Hard partition: $C_1 \cup \cdots \cup C_K = \{1, \dots, n\}$ and $C_k \cap C_{k'} = \emptyset$ for $k \neq k'$
- Two ingredients: how to measure similarity (a distance) and how to search for a good partition (an algorithm)
- Examples: customer segmentation, grouping cell lines by gene expression, grouping districts by socio-economic indicators

2 Distance and similarity measures

Dissimilarity and similarity

- A dissimilarity $d_{il} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) \geq 0$ is small when observations i and l are alike; clustering only needs the $n \times n$ matrix of pairwise dissimilarities
- A distance (metric) also satisfies, for any observations i, l and m: $d_{il} = 0 \iff \mathbf{x}^{(i)} = \mathbf{x}^{(l)}$, symmetry $d_{il} = d_{li}$, and the triangle inequality

$$d_{im} \leq d_{il} + d_{lm}$$

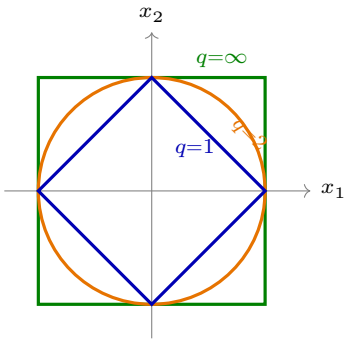
- A similarity s is the opposite (large for alike observations); if $s \in [0, 1]$, then $d = 1 - s$ is a dissimilarity
- The choice of d defines what "similar" means and can change the clusters more than the choice of algorithm

Minkowski distances

- For $\mathbf{x}^{(i)}, \mathbf{x}^{(l)} \in \mathbb{R}^p$,

$$d_q(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) = \Big(\sum_{j=1}^{p} |x_j^{(i)} - x_j^{(l)}|^q\Big)^{1/q}$$

- $q = 2$ Euclidean: the default; squaring gives large gaps more weight
- $q = 1$ Manhattan: sum of absolute differences; less dominated by one large gap
- $q \to \infty$ Chebyshev: $\max_j |x_j^{(i)} - x_j^{(l)}|$; only the worst coordinate counts



Points at distance 1 from the origin

Example data: USArrests

- Violent crime in the 50 US states in 1973
- No response and no group labels: a natural setting for unsupervised learning
- Variables are on different scales: three arrest rates per 100,000 residents and one percentage

Variable	Description	Mean	SD
Murder	murder arrests per 100,000 residents	7.79	4.36
Assault	assault arrests per 100,000 residents	170.76	83.34
UrbanPop	percent of the population living in urban areas	65.54	14.47
Rape	rape arrests per 100,000 residents	21.23	9.37

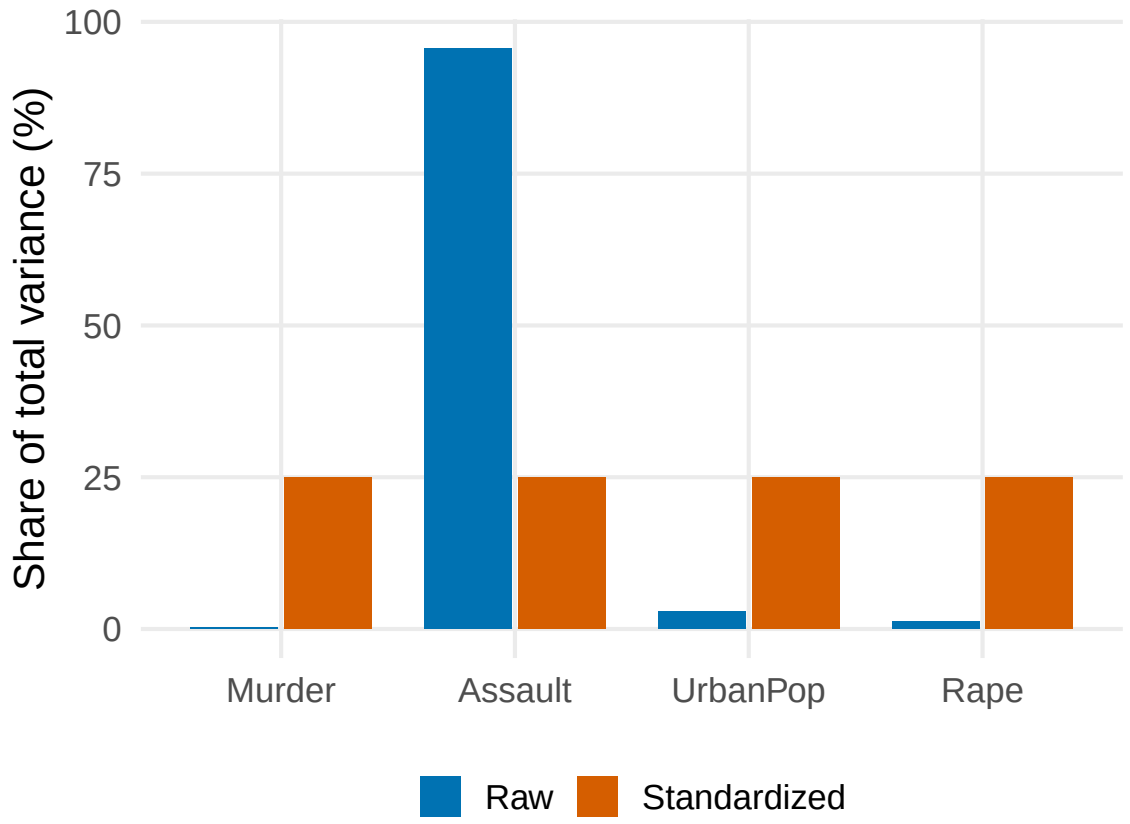
Sources: World Almanac and Book of Facts 1975 (arrests); Statistical Abstracts of the United States 1975 (urban population)

Scale matters: standardize first

- A variable measured on a large scale dominates the sum of squared (or absolute) differences
- Standardize each variable before computing distances:

$$z_j^{(i)} = \frac{x_j^{(i)} - \bar{x}_j}{\hat{\sigma}_j}$$

- Exception: variables already on a common, meaningful scale (e.g. counts of the same item)



USArrests: share of total variance

Mahalanobis distance

- Let $\mathbf{S}$ be the sample covariance matrix of the variables. Then

$$d_M(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) = \sqrt{(\mathbf{x}^{(i)} - \mathbf{x}^{(l)})^\top \mathbf{S}^{-1} (\mathbf{x}^{(i)} - \mathbf{x}^{(l)})}$$

- Rescales each variable and accounts for correlation between variables; equals Euclidean distance if $\mathbf{S} = \mathbf{I}$
- Use when variables are strongly correlated, so that Euclidean distance counts the same information several times
- Needs an invertible $\mathbf{S}$ (so $n > p$ and no exact collinearity), and $\mathbf{S}$ is estimated from all data, not within clusters

Correlation-based distance and cosine similarity

- Correlation between the profiles of observations i and l across the p variables, where $\bar{x}^{(i)}$ is the mean of observation i over its p variables:

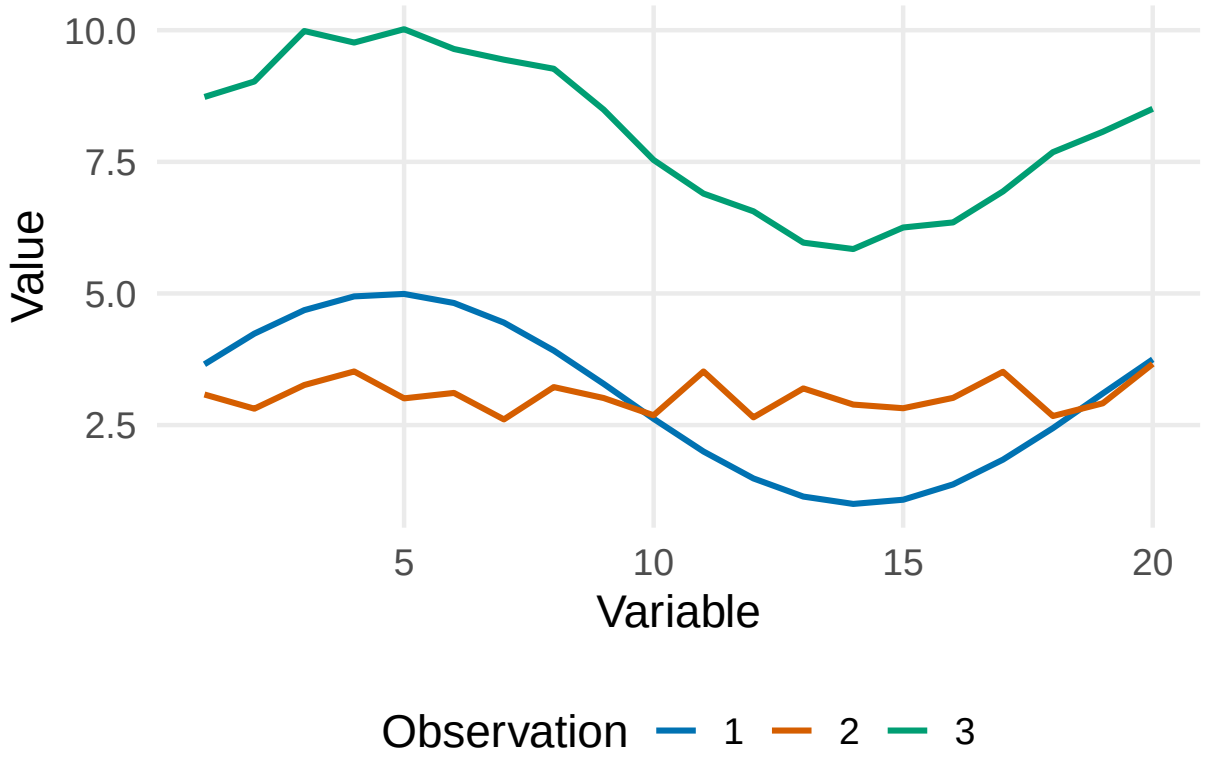
$$r_{il} = \frac{\sum_j (x_j^{(i)} - \bar{x}^{(i)})(x_j^{(l)} - \bar{x}^{(l)})}{\sqrt{\sum_j (x_j^{(i)} - \bar{x}^{(i)})^2} \sqrt{\sum_j (x_j^{(l)} - \bar{x}^{(l)})^2}}, \qquad d_{\text{cor}}(\mathbf{x}^{(i)}, \mathbf{x}^{(l)}) = 1 - r_{il}$$

- Cosine similarity is the same quantity without centering:

$$\cos\theta_{il} = \frac{\mathbf{x}^{(i)\top}\mathbf{x}^{(l)}}{\|\mathbf{x}^{(i)}\|\,\|\mathbf{x}^{(l)}\|}$$

When shape matters more than level

- Observations 1 and 3 have the same profile shape at different levels; observation 2 is flat
- Euclidean distance calls 1 and 2 close and 1 and 3 far apart; correlation distance says the opposite
- Use it when profile shape matters, not level (gene expression, shopping baskets, text); variables must be on comparable scales



Euclidean: d(1,2) = 6.1, d(1,3) = 22.4. Correlation distance: d(1,2) = 0.83, d(1,3) = 0.01

Simulated: 3 observations on 20 variables. 1 is a sine curve, 3 is observation 1 shifted up by 5, 2 is flat noise around 3.

Binary and categorical variables

- For two binary vectors, count the variables where they are $(1,1)$, $(1,0)$, $(0,1)$ and $(0,0)$: n_{11}, n_{10}, n_{01}, n_{00}, with $p = n_{11} + n_{10} + n_{01} + n_{00}$
- Simple matching: $s = \dfrac{n_{11} + n_{00}}{p}$
  - use when 0 and 1 are equally informative (symmetric), e.g. sex, urban/rural

- Jaccard: $s = \dfrac{n_{11}}{n_{11} + n_{10} + n_{01}}$
  - use when only presence is informative, e.g. rare symptoms, species present, items purchased
- Categorical variables with more than two levels: s is the fraction of variables on which the two observations agree; distance is $d = 1 - s$

Mixed data: Gower dissimilarity

- Survey data typically mix continuous, ordinal and categorical variables; Gower combines per-variable similarities

$$s_{il} = \frac{\sum_{j=1}^{p} w_j^{il}\, s_j^{il}}{\sum_{j=1}^{p} w_j^{il}}, \qquad d_{il} = 1 - s_{il}$$

- Continuous: $s_j^{il} = 1 - |x_j^{(i)} - x_j^{(l)}|/R_j$, where R_j is the range of variable j
- Categorical: $s_j^{il} = 1$ if $x_j^{(i)} = x_j^{(l)}$ and 0 otherwise; ordinal variables can use their ranks
- $w_j^{il} = 0$ if variable j is missing for i or l, else 1

Choosing a dissimilarity: summary

Measure	Variables	Use when
Euclidean	continuous	default; variables standardized, compact groups
Manhattan	continuous, counts	robustness to a single large difference
Mahalanobis	continuous	strongly correlated variables, $n > p$
Correlation, cosine	continuous	profile shape matters more than level
Matching	binary, categorical	0 and 1 equally informative
Jaccard	binary	only presence is informative
Gower	mixed	continuous, ordinal and categorical variables together

3 K-means clustering

K-means: the objective

- K is the number of clusters, chosen in advance; each cluster C_k is summarized by its mean (centroid), hence "K-means":

$$\bar{\mathbf{x}}_k = \frac{1}{|C_k|} \sum_{i \in C_k} \mathbf{x}^{(i)}$$

- Find the partition that minimizes the total within-cluster sum of squares

$$\min_{C_1, \ldots, C_K} \sum_{k=1}^{K} \sum_{i \in C_k} \left\| \mathbf{x}^{(i)} - \bar{\mathbf{x}}_k \right\|^2$$

- Equivalent (up to a factor of 2) to making pairwise squared distances within each cluster small: $\frac{1}{|C_k|} \sum_{i,l \in C_k} \|\mathbf{x}^{(i)} - \mathbf{x}^{(l)}\|^2$

K-means: the algorithm

There are almost $K^n/K!$ partitions: exhaustive search is impossible, so we need an algorithm.

1. Randomly assign each observation to one of K clusters
2. Repeat until the assignments stop changing:
   (a) compute the centroid $\bar{\mathbf{x}}_k$ of each cluster
   (b) assign each observation to the cluster with the closest centroid (Euclidean distance)

- This procedure is known as Lloyd's algorithm
- Each pass is cheap, $O(npK)$, so K-means scales to large n

Why the algorithm decreases the objective

- For any cluster C and any point $\mathbf{m}$,

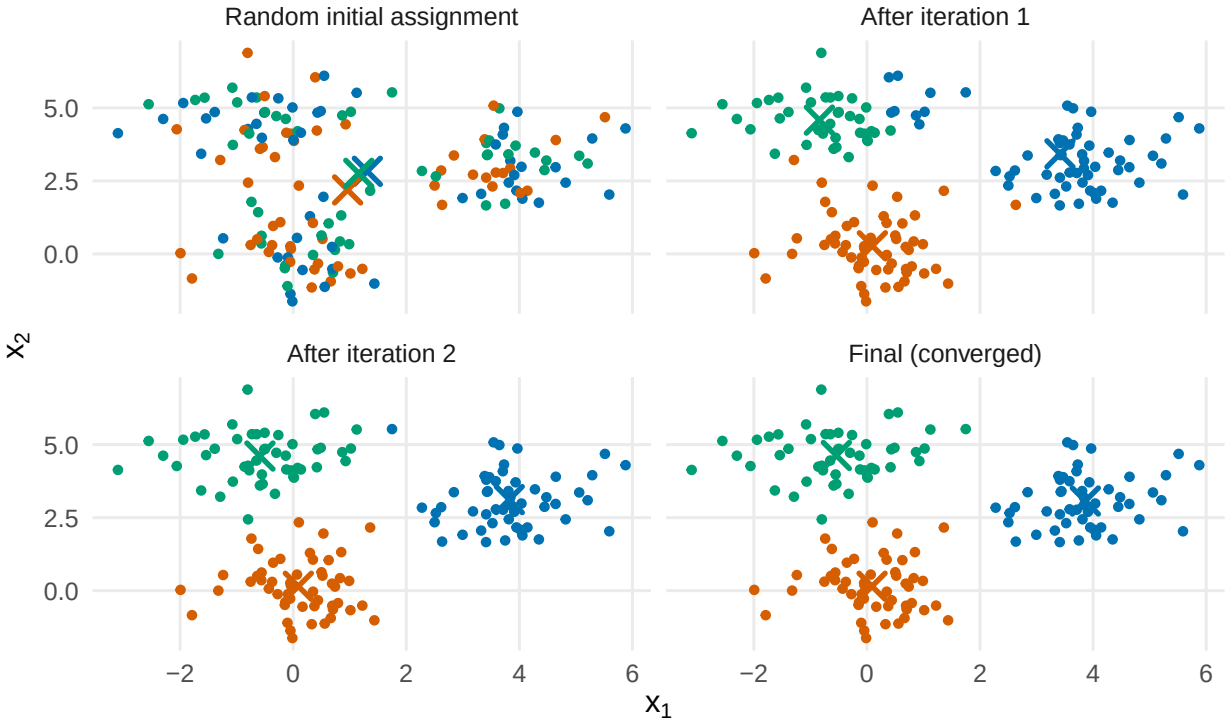
$$\sum_{i \in C} \|\mathbf{x}^{(i)} - \mathbf{m}\|^2 = \sum_{i \in C} \|\mathbf{x}^{(i)} - \bar{\mathbf{x}}_C\|^2 + |C| \, \|\bar{\mathbf{x}}_C - \mathbf{m}\|^2$$

- Step (a): the centroid is the best center for its cluster, so the objective cannot increase
- Step (b): moving an observation to its nearest centroid reduces its squared distance, so the objective cannot increase
- The objective is bounded below and there are finitely many partitions, so the algorithm converges, but only to a local optimum: a partition that neither step can improve, which need not be the best partition overall

Example data: simulated clusters

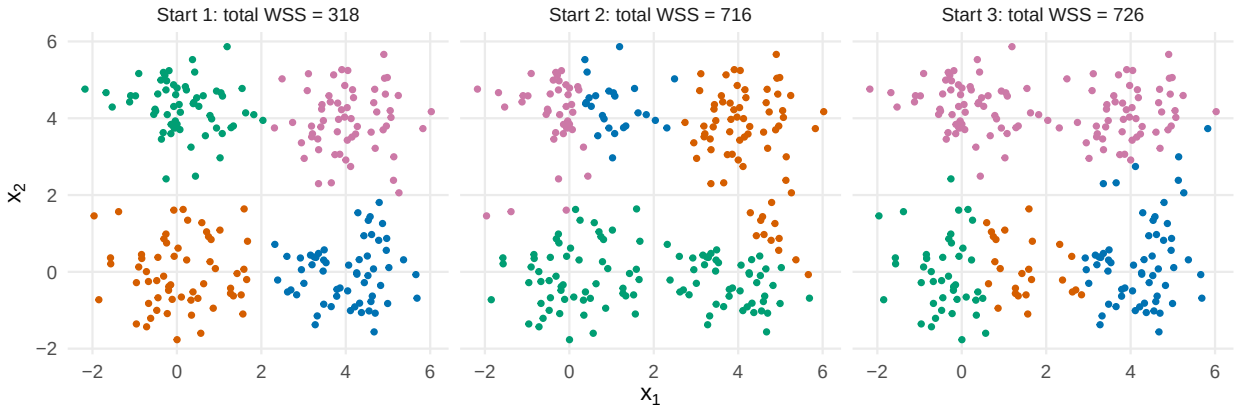
- The K-means examples, and some later ones, use simulated data with two variables, x_1 and x_2, so the true groups are known and can be plotted
- Each group is a round normal cloud: both coordinates are drawn independently from $N(\mu, \sigma^2)$ around the group's center
- Three-cluster data (K-means iterations): 3 groups of 50 observations, centers $(0, 0)$, $(4, 3)$ and $(-0.5, 4.5)$, $\sigma = 0.9$
- Four-cluster data (multiple starts, and later choosing K): 4 groups of 60 observations at the corners of a square, $(0, 0)$, $(4, 0)$, $(0, 4)$ and $(4, 4)$, $\sigma = 0.8$
- Smaller simulated sets used for single examples are described on the slides where they appear

K-means in action



Three-cluster simulated data, $K = 3$. Crosses mark the centroids.

Local optima and multiple starts



- Four-cluster data, $K = 4$, three single starts: the solution depends on the random initial assignment, and different starts can end in different local optima
- Run the algorithm from many random starts and keep the solution with the smallest objective
- Smarter initialization, k-means++: choose the initial centroids one at a time, each new one more likely to be a point far from those already chosen

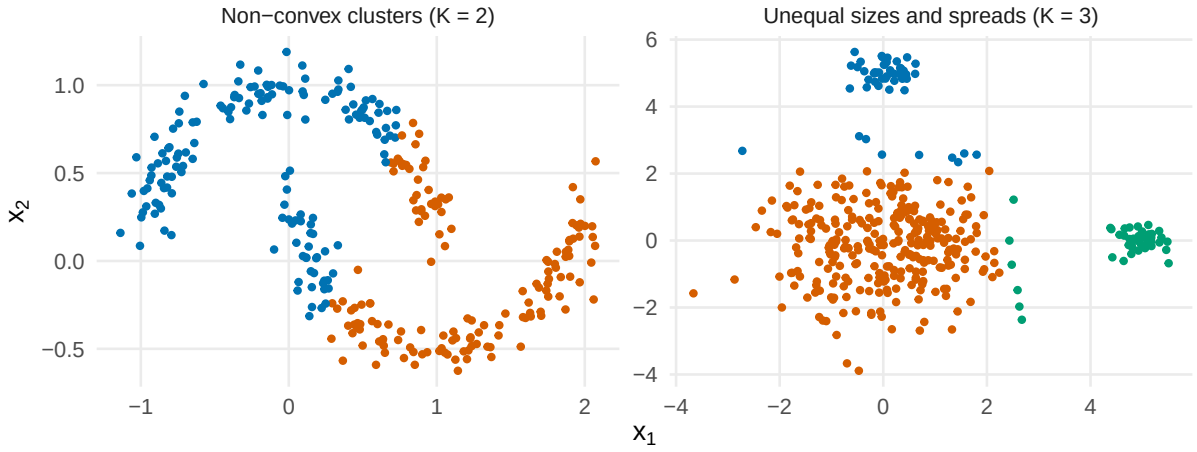
Choosing K and assigning new observations

- K must be specified before running the algorithm; the evaluation section covers how to choose it
- The objective always decreases as K grows, so it cannot be minimized over K
- A new observation $\mathbf{x}^*$ is assigned to the nearest centroid:

$$\hat{k}^* = \arg\min_k \|\mathbf{x}^* - \bar{\mathbf{x}}_k\|$$

- Standardize $\mathbf{x}^*$ using the means and standard deviations of the data used to fit the clusters

Strengths and limitations of K-means



Simulated. Left: two interleaved half-moons of 150 points each. Right: one large, spread-out group (300 points, $\sigma = 1.2$) and two small, tight groups (40 points each, $\sigma = 0.3$).

- Strengths: simple, fast, works well for compact, roughly spherical clusters of similar size
- Limitations: non-convex shapes (such as the half-moons), very unequal sizes or spreads, and outliers (centroids are means) can give poor clusters
- Also requires K, depends on the starting point, and assigns every observation to a cluster

4 Hierarchical clustering

Hierarchical clustering: the idea

- K-means needs K in advance; hierarchical clustering does not
- It builds a tree of nested clusters, the dendrogram, from which any number of clusters can be read off
- Agglomerative (bottom-up) is the most common: start with n singleton clusters and repeatedly fuse the two closest clusters
- The name comes from "agglomerate", to gather into a mass: clusters are built up by merging smaller ones

- Divisive (top-down) does the reverse: start with all n observations in one cluster and repeatedly split
- Needs a dissimilarity between observations and a rule for the dissimilarity between clusters (the linkage)

Agglomerative algorithm

1. Start with n clusters, one per observation, and compute all $\binom{n}{2}$ pairwise dissimilarities
2. For $m = n, n-1, \ldots, 2$:
   (a) find the pair among the m clusters with the smallest inter-cluster dissimilarity and fuse them; the dissimilarity is the height of the fusion in the dendrogram
   (b) recompute the inter-cluster dissimilarities among the remaining $m-1$ clusters

Divisive clustering

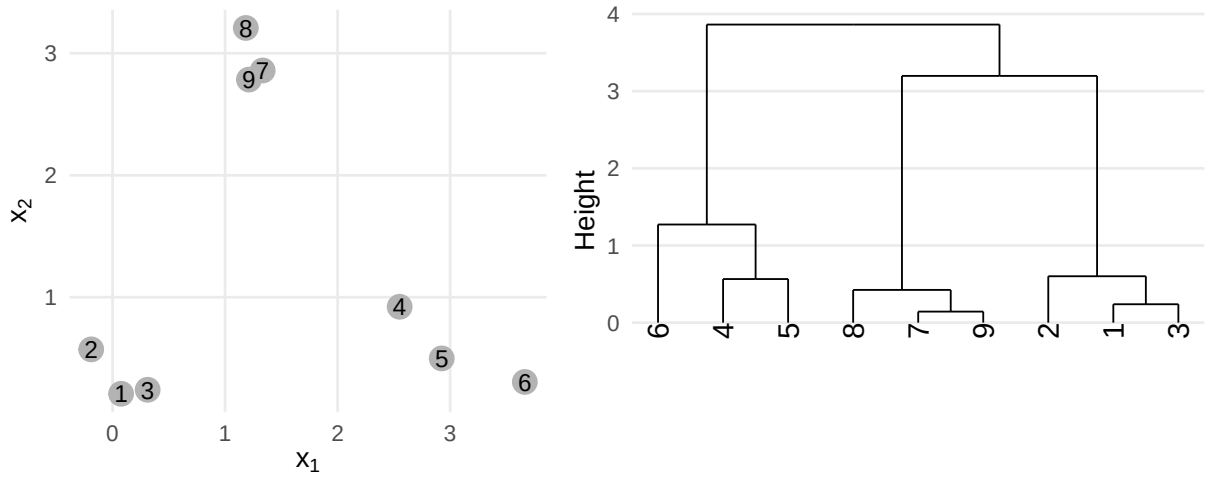
- Top-down: start with all n observations in one cluster and repeatedly split a cluster in two, until every observation is on its own
- A cluster C can be split in two in $2^{|C|-1}-1$ ways, too many to check, so methods use heuristics:
  - DIANA: split the cluster with the largest diameter (largest dissimilarity between two of its members); the member farthest on average from the rest starts a splinter group, and members closer on average to the splinter group move over
  - Bisecting K-means: split a cluster with K-means, $K = 2$
- In DIANA the height of a split is the diameter of the cluster split, so the result is again a dendrogram
- Settles the large clusters first, so it can do better when only a few big groups matter; less common in practice than agglomerative

Linkage: dissimilarity between clusters

Let A and B be two clusters and $d_{il} = d(\mathbf{x}^{(i)}, \mathbf{x}^{(l)})$.

- Complete: $d(A, B) = \max_{i \in A,\, l \in B} d_{il}$
- Single: $d(A, B) = \min_{i \in A,\, l \in B} d_{il}$
- Average: $d(A, B) = \dfrac{1}{|A|\,|B|} \sum_{i \in A} \sum_{l \in B} d_{il}$
- Centroid: $d(A, B) = d(\bar{\mathbf{x}}_A, \bar{\mathbf{x}}_B)$, the dissimilarity between the two cluster centroids
- Ward: fuse the pair of clusters whose merger increases the total within-cluster sum of squares the least (the K-means objective)

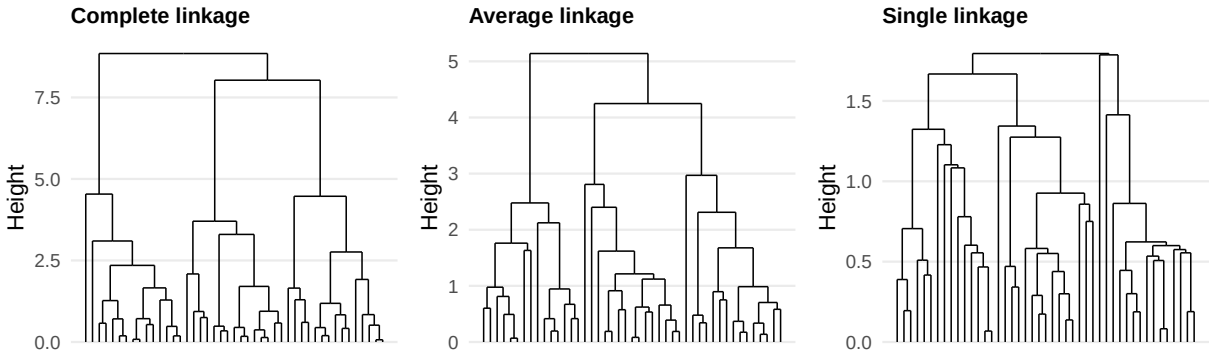
Reading a dendrogram



Simulated: 9 points in three groups of three; Euclidean distance, complete linkage

- Each leaf is an observation; the height of a fusion is the dissimilarity at which the two clusters joined
- Similarity of two observations is judged by the height where they first fuse, not by how close their leaves are horizontally
- Cutting the tree at a height gives a partition into clusters

Effect of linkage



Simulated: 45 points in three groups of 15 ($\sigma = 1$); Euclidean distance

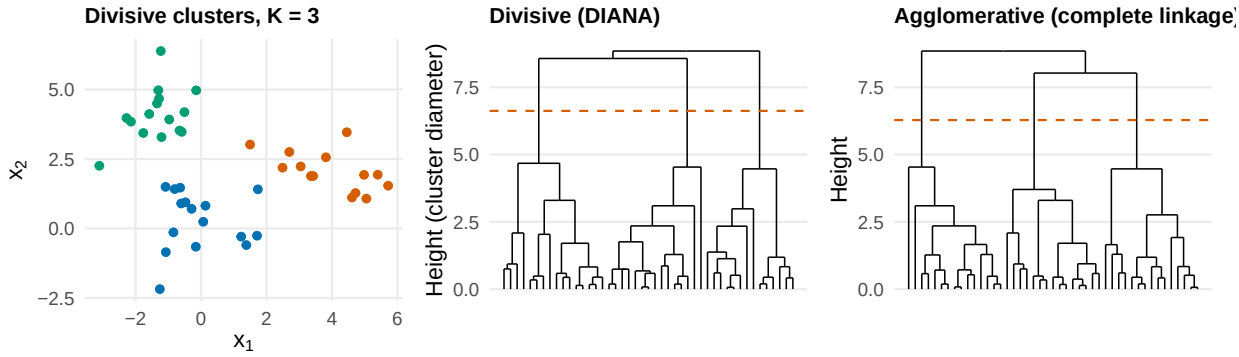
- Complete and average: compact, fairly balanced clusters; complete is the usual default
- Single: chaining, where clusters grow by adding one point at a time; useful for elongated groups, but gives trailing, unbalanced trees
- Centroid: popular in genomics, but can produce inversions (a fusion lower than an earlier one)

Cutting the tree

- Choose the number of clusters K or a height h at which to cut

- A large gap between successive fusion heights suggests a natural place to cut, but this is a judgment call
- Clusters are nested: those for $K = 2$ are merged clusters of $K = 3$
- Nesting can be unrealistic: a population split by sex and by nationality does not form a single hierarchy

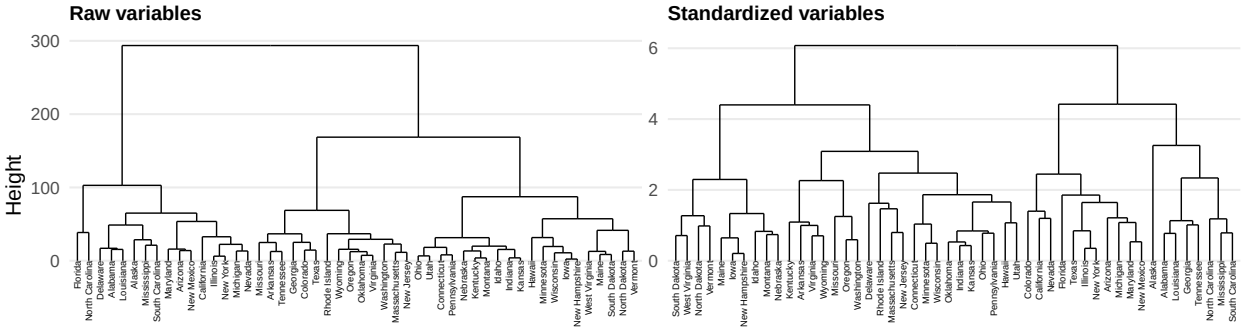
Divisive vs. agglomerative



Same 45 simulated points as the linkage comparison; Euclidean distance. Dashed lines cut each tree into $K = 3$ clusters.

- Left: the three clusters from the divisive (DIANA) tree; cutting the complete-linkage tree gives the same clusters except for one point
- Both trees are read the same way; in the divisive tree the height of a split is the diameter of the cluster being split

Effect of scaling on the clusters



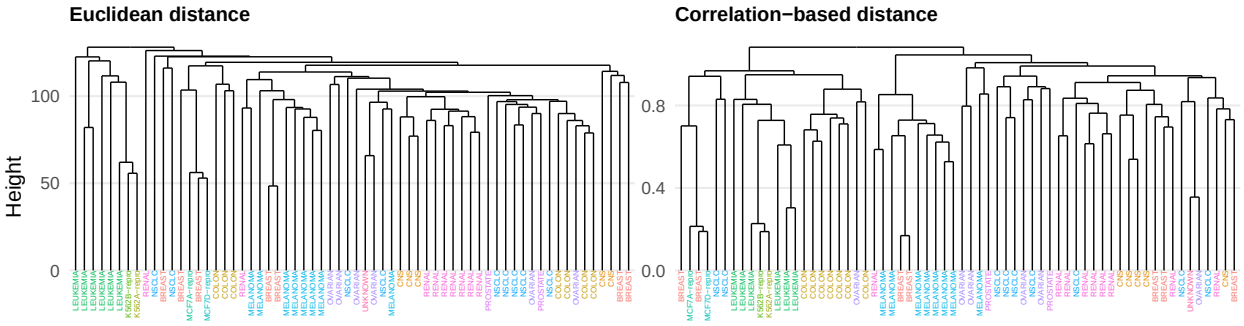
- USArrests, complete linkage: on raw data the tree is driven almost entirely by Assault, which has by far the largest variance
- After standardizing, all four variables contribute and the grouping of states changes

Example data: NCI60

- Gene expression for 64 human cancer cell lines (tumor cells grown in the laboratory) from the US National Cancer Institute's NCI-60 panel, measured with microarrays, which record the activity of thousands of genes at once (Ross et al., 2000)

- $p = 6{,}830$ genes per cell line, far more variables than observations ($p \gg n$), so Mahalanobis distance cannot be used; each gene is standardized before computing distances
- Each cell line carries a cancer-type label (14 labels):
  - breast 7, CNS 5, colon 7, leukemia 6, melanoma 8, non-small cell lung 9, ovarian 6, prostate 2, renal 9
  - 4 replicate samples (`K562A-repro`, `K562B-repro`, `MCF7A-repro`, `MCF7D-repro`) and 1 `UNKNOWN`, one cell line each
- The labels are not used to build the clusters; they are used afterwards to check whether the clusters match cancer types

Example: NCI60 gene expression



- Leaves are labeled and colored by cancer type, which was not used to build the tree
- Average linkage with Euclidean distance (left) and correlation-based distance between cell lines (right) group the cell lines differently

5 Cluster evaluation

Why evaluating clusters is hard

Without a response there is no ground truth, and every clustering algorithm returns clusters even when the data have no cluster structure.

Evaluation therefore asks several questions: how many clusters, how compact and well separated they are, whether they agree with known labels, and whether they are stable.

Internal measures use only the data and the partition; external measures compare with known labels; stability checks whether the result survives small changes.

Internal measure: within-cluster sum of squares

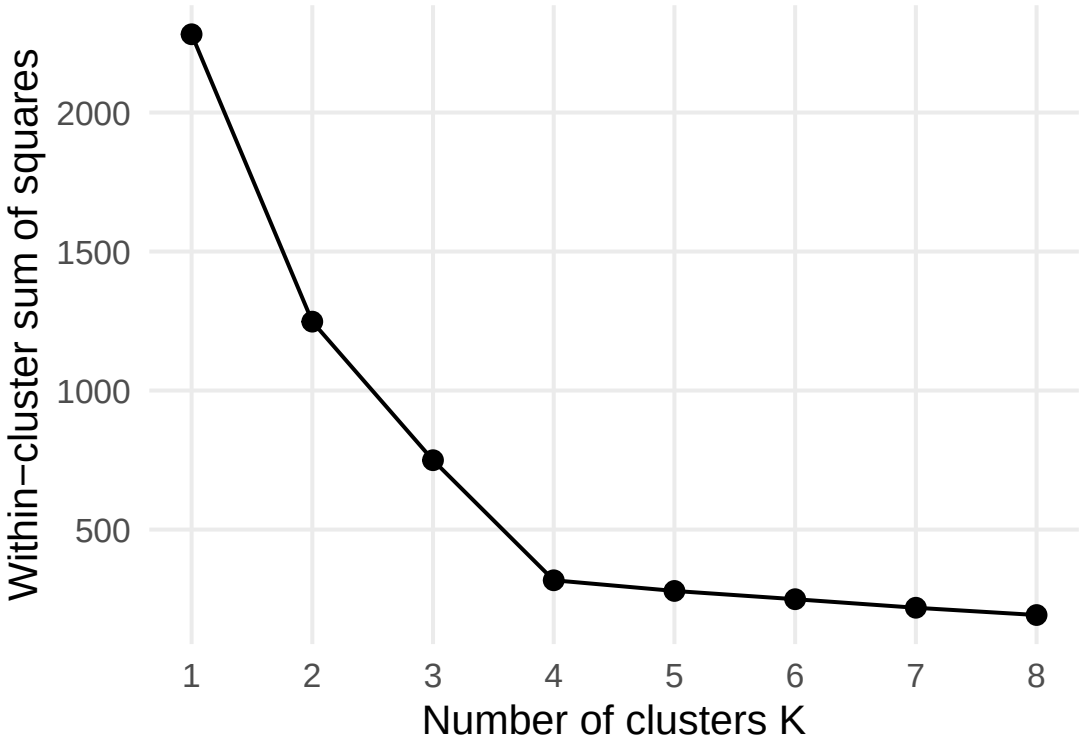
- Total variation: TSS = WSS + BSS ($\bar{\mathbf{x}}$: overall mean)

$$\text{WSS} = \sum_k \sum_{i \in C_k} \|\mathbf{x}^{(i)} - \bar{\mathbf{x}}_k\|^2$$

$$\text{BSS} = \sum_k |C_k| \, \|\bar{\mathbf{x}}_k - \bar{\mathbf{x}}\|^2$$

- BSS/TSS: share of variation explained by the clusters

- WSS always falls as K grows; look for an elbow, a K after which it falls much more slowly



Four-cluster simulated data. The elbow can be ambiguous.

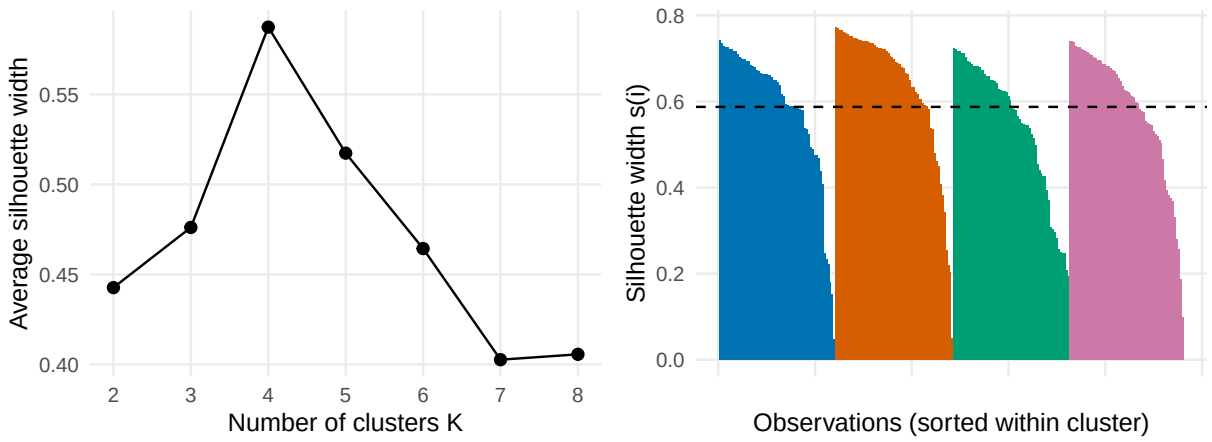
Internal measure: silhouette

- For observation i in cluster C_k, let $a(i)$ be its mean dissimilarity to the other members of C_k, and $b(i)$ the smallest mean dissimilarity to the members of any other cluster

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \in [-1, 1]$$

- $s(i)$ is the silhouette width of observation i; near 1: well clustered; near 0: on the border between clusters; negative: probably in the wrong cluster
- The average silhouette width summarizes a partition; choose the K that maximizes it

Silhouette in practice



- Four-cluster simulated data. Left: average silhouette width against K; right: individual silhouette widths for the chosen $K = 4$
- Unlike WSS, the average silhouette does not always improve with K, so it can point to a best K

Hierarchical clustering: cophenetic correlation

- The cophenetic distance t_{il} between observations i and l is the height at which they first join in the dendrogram
- The cophenetic correlation is the correlation between the original dissimilarities and the cophenetic distances over all pairs:

$$c = \mathrm{cor}(\{d_{il}\}_{i<l}, \{t_{il}\}_{i<l})$$

- Close to 1: the dendrogram represents the original dissimilarities faithfully
- Use it to compare linkages for the same dissimilarity

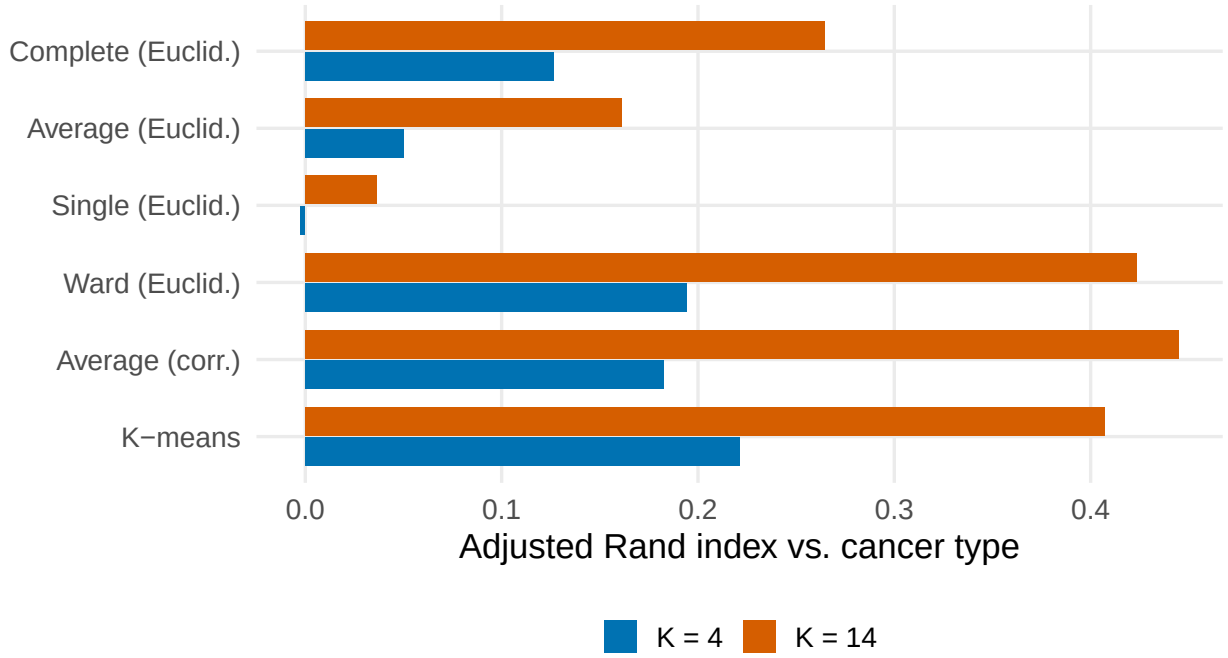
External measure: Rand index and adjusted Rand index

- When reference labels exist, compare the partition U with the labels V through pairs of observations
- Rand index: the fraction of pairs on which U and V agree, i.e. both place the pair together or both place it apart
- The adjusted Rand index corrects for agreement by chance. With n_{uv} the number of observations in cluster u of U and class v of V, $n_{u\cdot}$, $n_{\cdot v}$ the row and column totals, and E the value of $\sum_{u,v} \binom{n_{uv}}{2}$ expected by chance,

$$\mathrm{ARI} = \frac{\sum_{u,v} \binom{n_{uv}}{2} - E}{\frac{1}{2}\left(\sum_u \binom{n_{u\cdot}}{2} + \sum_v \binom{n_{\cdot v}}{2}\right) - E}, \qquad E = \frac{\sum_u \binom{n_{u\cdot}}{2} \sum_v \binom{n_{\cdot v}}{2}}{\binom{n}{2}}$$

- ARI $= 1$ for identical partitions and about 0 for random labeling (it can be negative)

NCI60: agreement with cancer type



- ARI between each clustering and the 14 cancer types, for $K = 4$ and $K = 14$ clusters
- Labels are not necessarily the true clusters: a low ARI may reflect a mismatch between the labels and the gene expression, not a bad clustering

Stability and robustness

- Clustering can be sensitive: removing a few observations or changing a setting may change the clusters
- Re-cluster on subsamples or bootstrap samples and check whether the same groups reappear
- Vary the choices: K, standardization, dissimilarity, linkage, and random starts
- Report conclusions that hold across reasonable choices, and treat the rest as uncertain

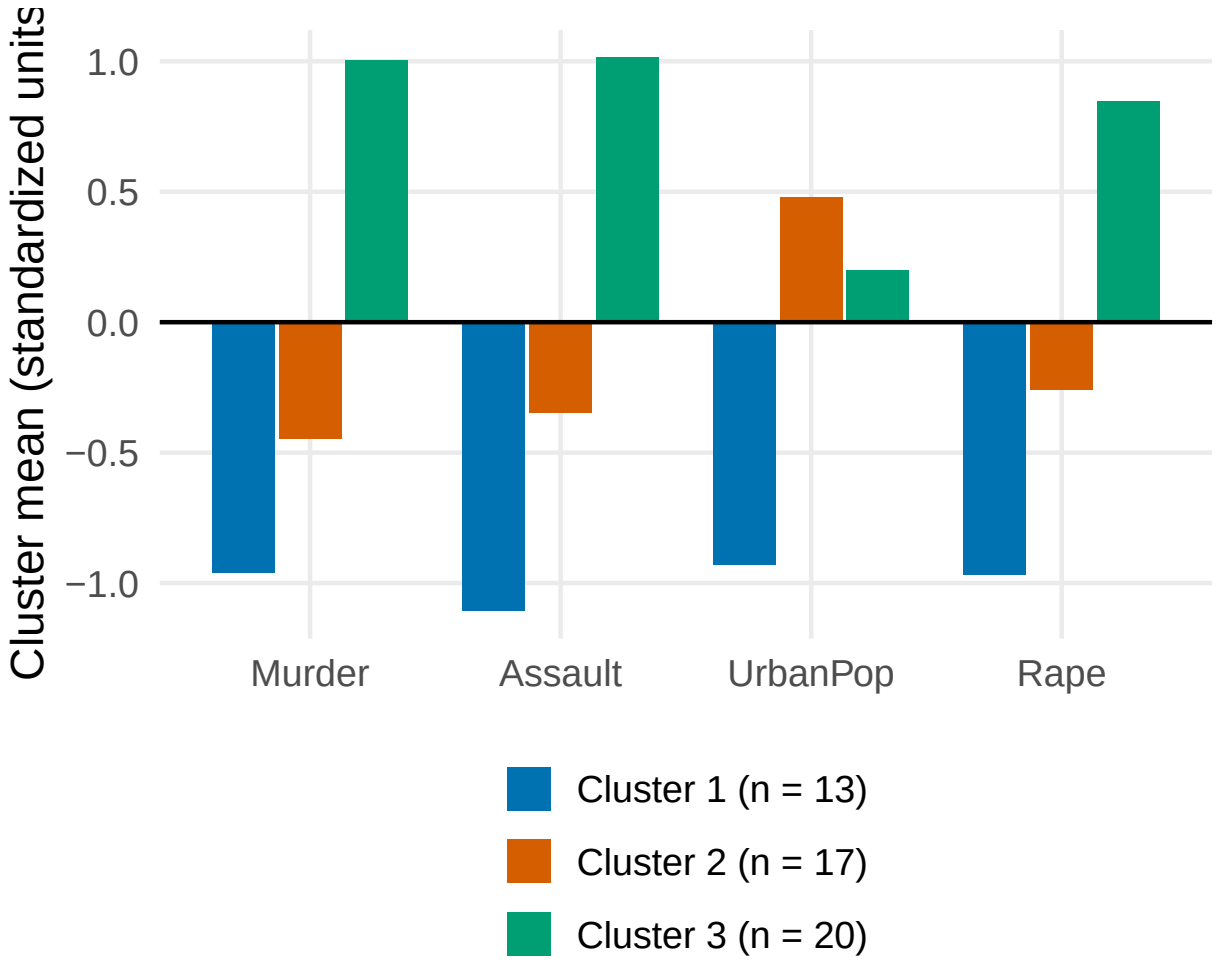
Evaluation tools: summary

Tool	Needs labels?	Answers
WSS and elbow	no	how much variation K clusters explain; choosing K
Silhouette	no	how well separated and compact; choosing K
Cophenetic correlation	no	how well a dendrogram preserves the dissimilarities
Adjusted Rand index	yes	agreement with a reference partition
Stability (resampling)	no	whether the clusters are reproducible

6 Cluster interpretation

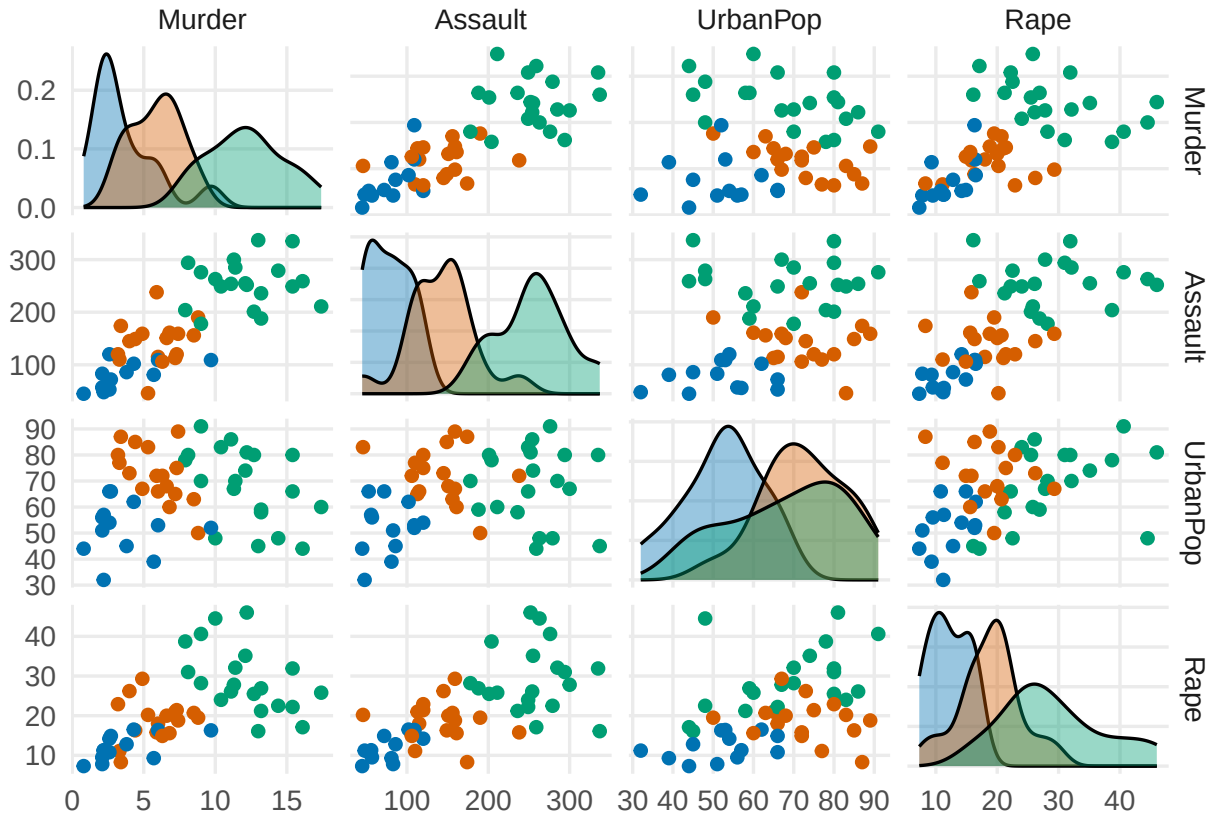
Profiling the clusters

- Report each cluster's size $|C_k|$ and centroid $\bar{\mathbf{x}}_k$ in original units
- Compare clusters on a common scale with standardized means $(\bar{x}_{kj} - \bar{x}_j)/\hat{\sigma}_j$; $\bar{x}_{kj}$ is the mean of variable j in cluster k
- Variable-by-variable tests (ANOVA) are descriptive only: the clusters were chosen to differ, so p-values are not valid



USArrests, K-means with $K = 3$

Visualizing clusters: pairs plot



- Every pair of the original variables, colored by cluster: shows directly which variables separate which clusters

Interpreting and reporting clusters

Name clusters from their profiles, but remember that a cluster is a description of this sample under one set of choices, not a discovered ground truth.

Report the dissimilarity, standardization, algorithm, K, number of starts, and the evidence from evaluation and stability.

Check the clusters with subject-matter knowledge or new data; do not use the same variables to test whether the clusters differ.

Cluster labels can be used downstream, for example as a feature in a supervised model.

Summary

- Clustering partitions unlabeled observations; the result depends on the dissimilarity, scaling, algorithm and K
- Standardize unless variables share a meaningful scale; correlation distance for profile shape, Gower for mixed data
- K-means minimizes within-cluster sum of squares with Lloyd iterations, so use many starts; hierarchical clustering gives a dendrogram for any K, and the linkage matters
- Evaluate with elbow and silhouette for K, cophenetic correlation for dendrograms, ARI when labels exist, and stability always

- Interpret with cluster profiles and visualizations; reading: ISLP Chapter 12, Section 12.4